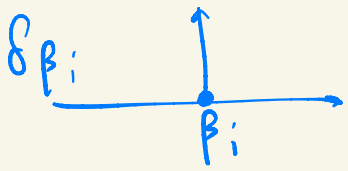
Week 10 Leave-one-out Method

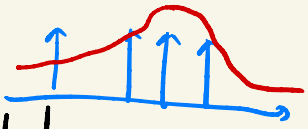
- Empirical distribution : $\hat{\beta} \in \mathbb{R}^p$

$$\Rightarrow \hat{P}_{\hat{\beta}} = \sum_{i=1}^{p} \delta_{\hat{\beta}_i}$$



$\hat{P}_{\hat{\beta}} \xrightarrow{\text{weakly}} P$ means :

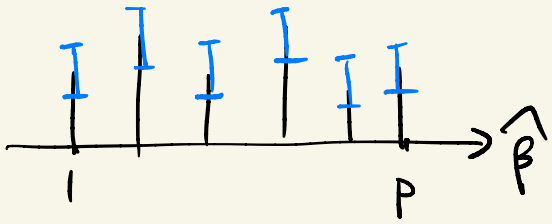
$$\int f \hat{P}_{\hat{\beta}} \xrightarrow{p \to \infty} \int f P \,, \forall f \text{ bounded continuous function}$$



$$\|\hat{\beta}\|_2^2 = \int_{x=-\infty}^{+\infty} x^2 \hat{P}_{\hat{\beta}} \,,$$ so convergence of empirical distribution implies convergence of $\|\hat{\beta}\|$.

- Applications :

Constructing Confidence interval



$y = A\beta + \varepsilon$

$\rightarrow \hat{\beta} = \text{function}(y, A)$

If I know $\hat{P}_{\hat{\beta}-\beta}$ is $N(0, \sigma^2)$

$\Rightarrow \hat{\beta}_i - \beta_i \sim N(0, \sigma^2)$, $\forall i$

$\Rightarrow$ Construct a CI of the form

$$\left[\hat{\beta}_i - \sigma u_{\alpha/2}, \hat{\beta}_i + \sigma u_{\alpha/2}\right]$$

- This lecture:

  leave-one-out technique (signal processing, optimization)

  Cavity method, Thouless-Anderson-Palmer (TAP)

  Coordinate descent (optimization)

  predecessor comparison (random matrix theory)

- A simple example

  A: $n \times p$ matrix, iid $\mathcal{N}(0,1)$ entries

  $n/p = \delta > 1$, $n, p \to \infty$

  $w \in \mathbb{R}^n$, iid $\mathcal{N}(0, n)$

  $y = w + A\theta$

  $\hat{\theta} - \theta := \underset{\theta \in \mathbb{R}^d}{\operatorname{argmin}} \|y - A\theta\|_2$

  $= (A^T A)^{-1} A^T w$

  $\Rightarrow$ We can assume W.L.O.G. $\theta = 0$, so $y = w$

  Goal: $\hat{P}_{\hat{\theta}} \xrightarrow{p \to \infty} \mathcal{N}\left(0, \frac{\delta}{\delta - 1}\right)$

First: note the rotation invariance: $\forall Q$ orthogonal,

$$Q\hat{\theta} = Q(A^TA)^{-1}A^Tw$$

$$= (A^TAQ^T)^{-1}A^Tw$$

$$= (QA^TAQ^T)^{-1}QA^Tw$$

$$\overset{\text{distribution}}{=} (A^TA)^{-1}A^Tw$$

$$= \hat{\theta}$$

$$\Rightarrow \hat{\theta} \overset{\text{distribution}}{=} \|\hat{\theta}\|_2 \, u$$

u: uniform on S^{p-1}

$$\approx \mathbb{E}[\|\hat{\theta}\|_2]\, u$$

$$\hat{P}_{\sqrt{p}\,u} \longrightarrow N(0,1)$$

But, finding $\mathbb{E}\|\hat{\theta}\|_2$ is nontrivial!

- Approach from random matrix theory

Marchenko–Pastur law:

empirical distribution of eigenvals of $\frac{1}{n}A^TA$ $\xrightarrow{p\to\infty}$

$$\mu(dx) = \frac{1}{2\pi}\cdot\frac{\sqrt{(\lambda_+ - x)(x-\lambda_-)}}{\lambda x}\,dx, \quad \forall \lambda_- \le x \le \lambda_+$$

$$\lambda := \delta^{-1} < 1, \quad \lambda_\pm := (1 \pm \sqrt{\lambda})^2$$

$$\frac{1}{p}\mathbb{E}\left[\hat{\theta}^T\hat{\theta}\,\middle|\,A\right] = \frac{1}{p}\operatorname{tr}\left(\left(\frac{1}{n}A^TA\right)^{-1}\right)$$

$$= \int \frac{1}{x}\,\mu_p(dx)$$

μ_p → empirical measure of eigenvals of $\frac{1}{n}A^TA$

$$\xrightarrow{p\to\infty} \int \frac{1}{x}\,\mu(dx)\,\frac{1}{2\pi}\cdot\frac{\sqrt{(\ )(\ )}}{\lambda x}\,dx$$

$t = \frac{(1-\lambda)^2}{x}$

$$= \frac{\delta}{\delta-1}\int \mu(dt)\to \frac{1}{2\pi}\cdot\frac{\sqrt{(\ )(\ )}}{\lambda t}\,dt$$

$$dt = -\frac{(1-\lambda)^2}{x^2}\,dx$$

$$= \frac{\delta}{\delta-1}$$

- A general optimization problem:

$$\varepsilon := \frac{1}{\sqrt{n}}w \in \mathbb{R}^n$$

$$\hat{\beta} := \operatorname*{argmin}_{\beta}\sum_{i=1}^{n}\rho(\varepsilon_i - A_i^T\beta)$$

$\rho: \mathbb{R}\to[0,\infty)$ convex.

$A = \begin{bmatrix} A_1^T \\ \vdots \\ A_n^T \end{bmatrix}$, $\rho(t) = t^2 \Rightarrow \hat{\theta} = \sqrt{n}\hat{\beta}$

- $\text{prox}_c(\rho)(x) := \underset{y}{\text{argmin}} \left\{ \rho(y) + \frac{1}{2c}(x-y)^2 \right\}$

Thm: Let $\varepsilon \sim N(0,1)$ be scalar, $Z \sim N(0, r^2) + \varepsilon$, $r := \lim_{p\to\infty} \|\hat{\beta}\|$. Then, $\exists\, c > 0$ s.t.

$$\begin{cases} \mathbb{E}\left[(\text{prox}_{c(\rho)})'(Z)\right] = 1 - \delta^{-1} \\ \delta^{-1} r^2 = \mathbb{E}\left[(Z - \text{prox}_{c(\rho)}(Z))^2\right] \end{cases}$$

solve c, and $\|\hat{\beta}\|$

Take $\rho(x) = x^2$,

$$\text{Prox}_{c(\rho)}(x) = \frac{x}{2c+1}$$

$$Z \sim N(0, 1+r^2)$$

$$\Rightarrow \begin{cases} \frac{1}{2c+1} = 1 - \delta^{-1} \\ \delta^{-1} r^2 = \left(\frac{2c}{2c+1}\right)^2 (1+r^2) \end{cases}$$

$$\Rightarrow \quad r^2 = \frac{1}{\delta - 1}$$

$$\Rightarrow \quad \hat{P}_{\sqrt{n}\hat{\beta}} = N\left(0, \frac{n}{p} \cdot r^2\right) = N\left(0, \frac{\delta}{\delta - 1}\right).$$

- Normal Equation

Define

$$\psi(x) := \rho'(x)$$

$$\Rightarrow \quad \sum A_i \psi(\varepsilon_i - A_i^T \hat{\beta}) = 0$$

Residue: $R_i = \varepsilon_i - A_i^T \hat{\beta}$

- Leaving out an Observation

$\hat{\beta}_{(i)}$: leave-one-out estimator

$$\sum_{j \neq i} A_j \psi(\varepsilon_j - A_j^T \hat{\beta}_{(i)}) = 0$$

Define

$$r_{j(i)} := \varepsilon_j - A_j^T \hat{\beta}_{(i)}$$

Take difference

$$\Rightarrow A_i \psi(\varepsilon_i - A_i^T \hat{\beta}) + \underbrace{\sum_{j \neq i} \psi'(r_{j,(i)}) A_j A_j^T}_{S_i \text{ def}} (\hat{\beta}_{(i)} - \hat{\beta}) \approx 0$$

$$\Rightarrow \quad R_i - r_{i,(i)} = -A_i^T\left(\hat{\beta} - \hat{\beta}_{(i)}\right)$$

$$\approx -A_i^T S_i^{-1} A_i \psi(R_i)$$

$$\approx -\mathrm{tr}(S_i^{-1})\psi(R_i)$$

If $B \in \mathbb{R}^{p\times p}$, $u \sim N(0, I_p)$

$\Rightarrow u^T B u \approx \mathrm{Tr}(B)$ w.h.p. (measure concentration)

By symmetry, assume that

$$c \approx \mathrm{tr}\left(S_i^{-1}\right), \quad i = 1, \cdots, n$$

$$\boxed{R_i - r_{i,(i)} \approx -c\psi(R_i)}$$

$\rightarrow r_{i,(i)} \approx R_i + c\psi(R_i)$

$\Rightarrow R_i = \mathrm{prox}_{c\rho}(r_{i,(i)})$

- Leaving out p-th Predicator.

$$A_i = \begin{bmatrix} V_i \\ A_{ip} \end{bmatrix}, \quad V_i \in \mathbb{R}^{p-1}$$

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_{\cdot p} \\ \hat{\beta}_p \end{bmatrix}, \quad \hat{\beta}_{\cdot p} \in \mathbb{R}^{p-1}$$

$\hat{\gamma}$: leave-one-out estimator

$$\sum_i V_i \psi\left(\underbrace{\varepsilon_i - V_i^T \hat{\gamma}}_{= r_{i[p]}}\right) = 0$$

Taking difference,

$$\begin{cases} \sum_i V_i \left[\psi(R_i) - \psi(r_{i[p]})\right] = 0_{p-1} \\ \sum_i A_i(p)\, \psi(R_i) = 0 \end{cases}$$

Defn $\Rightarrow$

$$R_i - r_{i[p]} = V_i^T(\hat{\gamma} - \hat{\beta}_{\setminus p}) - A_i(p)\hat{\beta}_p$$

Taylor $\Rightarrow$

$$\begin{cases} \left[\underbrace{\sum_i \psi'(r_{i[p]}) V_i V_i^T}_{=: G_p}\right](\hat{\gamma} - \hat{\beta}_{\setminus p}) - \hat{\beta}_p \underbrace{\sum_i \psi'(r_{i[p]}) V_i A_i(p)}_{=: u_p} \approx 0_{p-1} \\ \sum_i A_i(p)\left[\psi(r_{i[p]}) + \psi'(r_{i[p]})\left(V_i^T(\hat{\gamma} - \hat{\beta}_{\setminus p}) - A_i(p)\hat{\beta}_p\right)\right] \approx 0 \end{cases}$$

$$\Rightarrow \hat{\beta}_p \approx \frac{\sum_i A_i(p)\psi(r_{i[p]})}{\sum_i A_i^2(p)\psi'(r_{i[p]}) - u_p^T G_p^{-1} u_p}$$

$$D := \mathrm{diag}\left(\left[\psi'(r_{i(p)})\right]_{i=1}^{n}\right)$$

$$A(p) := \begin{bmatrix} A_1(p) \\ \vdots \\ A_n(p) \end{bmatrix} \in \mathbb{R}^n$$

$$V = \begin{bmatrix} V_1^T \\ \vdots \\ V_n^T \end{bmatrix} \in \mathbb{R}^{n \times (p-1)}$$

$$u_p^T G_p^{-1} u_p = [A(p)^T D V]\,[V^T D V]^{-1}\,[V^T D A(p)]$$

$$\approx \mathrm{tr}\left(D D^{\frac{1}{2}} V (V^T D V)^{-1} V^T D^{\frac{1}{2}}\right)$$

$$= \mathrm{tr}(DP)$$

Defn projection matrix

$$= \sum_i D_{ii} P_{ii}$$

$$\Rightarrow \hat{\beta}_p \approx \frac{\sum_i A_i(p)\psi(r_{i(p)})}{\sum_i A_i(p)^2 D_{ii}(1-p_{ii})} \qquad (*)$$

measure concentration

$$\sum_i A_i(p)^2 D_{ii}(1-p_{ii}) \approx \mathbb{E}[\cdots] = \sum_i D_{ii}(1-p_{ii})$$

$$P_{ii} = D_{ii}\, V_i^T G_p^{-1} V_i$$

Defn

$$= D_{ii}\, V_i^T \left(G_{p(i)} + D_{ii} V_i V_i^T \right)^{-1} V_i$$

$G_{p(i)} = \sum_{j \neq i} D_{jj} V_j V_j^T$

$G_{p(i)}$ → independent of V_i !

Sherman–Morrison

$$\Rightarrow \left(G_{p(i)} + D_{ii} V_i V_i^T \right)^{-1}$$

$$= G_{p(i)}^{-1} - \frac{D_{ii} G_{p(i)}^{-1} V_i V_i^T G_{p(i)}^{-1}}{1 + D_{ii} V_i^T G_{p(i)}^{-1} V_i}$$

→ smaller in trace, but coherent with V_i !

$$\Rightarrow 1 - P_{ii} = \frac{1}{1 + D_{ii} V_i^T G_{p(i)}^{-1} V_i}$$

$$\approx \mathrm{tr}\left(G_{p(i)}^{-1}\right)$$

$$\approx \mathrm{tr}\left(G_p^{-1}\right)$$

$$\approx \mathrm{tr}\left(S_i^{-1}\right)$$

$$\approx C$$

$$\Rightarrow \boxed{\sum_{i=1}^{p} \frac{1}{1 + D_{ii} C} \approx \sum (1 - P_{ii}) = n - \mathrm{tr}(P) = n - (p-1) \approx n - p}$$

$D_{ii} = \psi'(R_i)$

Denominator in (*):

$$\sum_i A_i(\rho)^2 D_{ii}(1-\rho_{ii}) \approx \sum_i D_{ii}(1-\rho_{ii})$$

$$\approx \sum_i \frac{D_{ii}}{1+cD_{ii}}$$

$$\approx \frac{1}{c}\sum_i \rho_{ii}$$

$$\approx \frac{p}{c}$$

$$\Rightarrow \quad \hat{\beta}_\rho \approx \frac{c}{p}\sum_i A_i(\rho)\psi(r_{i(\rho)})$$

$$\Rightarrow \mathbb{E}\left[\hat{\beta}_\rho^2 \mid V, c\right] = \frac{c^2}{p^2}\sum_i \psi^2(r_{i(\rho)})$$

$$\approx \frac{c^2}{p^2}\sum_i \psi(R_i)$$

$$\boxed{\mathbb{E}[\|\beta\|^2] \approx \frac{c^2}{p}\sum \psi^2(R_i)}$$

but $r_{i(i)} = \varepsilon_i - A_i^T\hat{\beta}_{(i)} \sim \mathcal{N}(0,\ 1+\|\hat{\beta}\|^2)$

we can solve 3 independent objects, R_i, $\|\hat{\beta}\|$, c from the three equations in boxes.

- Using prox_c, we can rewrite those as the 2 fixed point equations (see pdf notes)

Summary of key ideas:

- Taylor expansion
- correlation in inner products

  e.g. $g : \mathbb{R}^p \to \mathbb{R}^p$, coordinatewise function, $\varepsilon = \frac{1}{\sqrt{p}}$

  $$g(\varepsilon\Delta) \approx g(0) + \underbrace{\nabla g(0) \cdot \varepsilon\Delta}_{\text{small in norm}}$$

  but $$\Delta^T g(\varepsilon\Delta) \approx \Delta^T g(0) + \underbrace{\varepsilon \Delta^T \nabla g(0) \Delta}_{\text{Comparable}}$$

- Concentration of measure